

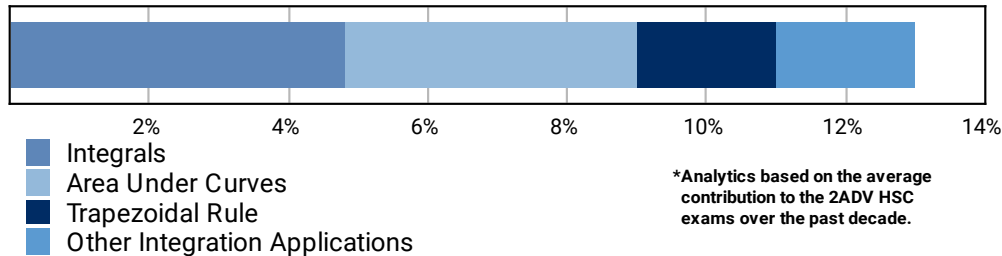
ADV: Calculus (Adv), C4 Integration (Adv), Integration (Y12)

Trig Integration

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Exam Equivalent Time: 42 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C4 Integration



HISTORICAL CONTRIBUTION

- C4 Integration has contributed a substantial 13.0% to past Advanced exams, on average over the past decade.
- This topic has been split into four sub-topics for analysis purposes: 1-Integrals (4.8%), 2-Areas Under Curves (4.2%), 3-Trapezoidal Rule (2.0%) and 4-Other Integration Examples (2.0%).
- This analysis looks at Integrals.

HSC ANALYSIS - What to expect and common pitfalls

- Integrals (4.8%) is made up of standard integration (1.2%), log and exponential integration (2.4%) and trig integration (1.2%).
- Dedicated questions testing trig integration were most recently asked in 2019 and 2020.
- Trig Integration is typically tested at the band 4 difficulty level and gives students a great opportunity for scoring highly.
- Two-step questions that require students to differentiate trig functions and then use that information to integrate have been worth 4 full marks in the past and deserve attention (asked twice since 2008).

Questions

1. Calculus, 2ADV C4 2019 HSC 9 MC

Which expression is equal to $\int \tan^2 x \, dx$?

- (A) $\tan x - x + C$
(B) $\tan x - 1 + C$
(C) $\frac{\tan^3 x^2}{6} + C$
(D) $\frac{\tan^3 x}{3} + C$

2. Calculus, 2ADV C4 2020 HSC 13

Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$. (2 marks)

3. Calculus, 2ADV C4 2008 HSC 3b

i. Differentiate $\log_e(\cos x)$ with respect to x . (2 marks)

ii. Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$. (2 marks)

4. Calculus, 2ADV C4 2007 HSC 2bi

Find $\int (1 + \cos 3x) \, dx$. (2 marks)

5. Calculus, 2ADV C4 2008 HSC 5a

The gradient of a curve is given by $\frac{dy}{dx} = 1 - 6 \sin 3x$. The curve passes through the point (0, 7).

What is the equation of the curve? (3 marks)

6. Calculus, 2ADV C4 2012 HSC 11g

Find $\int_0^{\frac{\pi}{2}} \sec^2\left(\frac{x}{2}\right) \, dx$ (3 marks)

7. Calculus, 2ADV C4 2014 HSC 11e

Evaluate $\int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) dx$. (3 marks)

8. Calculus, 2ADV C4 2014 HSC 13a

i. Differentiate $3 + \sin 2x$. (1 mark)

ii. Hence, or otherwise, find $\int \frac{\cos 2x}{3 + \sin 2x} dx$. (2 marks)

9. Calculus, 2ADV C4 2008 HSC 2cii

Evaluate $\int_0^{\frac{\pi}{12}} \sec^2 3x dx$. (3 marks)

10. Calculus, 2ADV C4 2010 HSC 5b

i. Prove that $\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x}$. (1 mark)

ii. Hence prove that $\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}$. (1 mark)

iii. Hence, use the identity $\int \sec ax \tan ax dx = \frac{1}{a} \sec ax$ to find the exact value of

$$\int_0^{\frac{\pi}{4}} \frac{1}{1 - \sin x} dx. \quad (2 \text{ marks})$$

Worked Solutions

1. Calculus, 2ADV C4 2019 HSC 9 MC

Consider option A:

$$\frac{d}{dx}(\tan x - x + C)$$

$$= \sec^2 x - 1$$

$$= \tan^2 x$$

$$\therefore \int \tan^2 x dx = \tan x - x + C$$

$$\Rightarrow A$$

♦♦ Mean mark 35%.

2. Calculus, 2ADV C4 2020 HSC 13

$$\int_0^{\frac{\pi}{4}} \sec^2 x = [\tan x]_0^{\frac{\pi}{4}}$$

$$= \tan \frac{\pi}{4} - \tan 0$$

$$= 1$$

3. Calculus, 2ADV C4 2008 HSC 3b

i. $y = \log_e(\cos x)$

$$\begin{aligned}\frac{dy}{dx} &= \frac{-\sin x}{\cos x} \\ &= -\tan x\end{aligned}$$

ii. $\int_0^{\frac{\pi}{4}} \tan x \, dx$

$$\begin{aligned}&= -[\log_e(\cos x)]_0^{\frac{\pi}{4}} \\ &= -\left[\log_e\left(\cos\left(\frac{\pi}{4}\right)\right) - \log_e(\cos 0)\right] \\ &= -\left[\log_e\left(\frac{1}{\sqrt{2}}\right) - \log_e 1\right] \\ &= -\left[\log_e\left(\frac{1}{\sqrt{2}}\right) - 0\right] \\ &= -\log_e\left(\frac{1}{\sqrt{2}}\right) \\ &= 0.346... \\ &= 0.35 \text{ (2 d.p.)}\end{aligned}$$

4. Calculus, 2ADV C4 2007 HSC 2bi

$$\begin{aligned}\int (1 + \cos 3x) \, dx \\ = x + \frac{1}{3}\sin 3x + C\end{aligned}$$

5. Calculus, 2ADV C4 2008 HSC 5a

$$\frac{dy}{dx} = 1 - 6 \sin 3x$$

$$\begin{aligned}y &= \int 1 - 6 \sin 3x \, dx \\ &= x + 2 \cos 3x + c\end{aligned}$$

Passes through (0, 7)

$$\Rightarrow 0 + 2 \cos 0 + c = 7$$

$$2 + c = 7$$

$$c = 5$$

$\therefore$ Equation is $y = x + 2 \cos 3x + 5$

6. Calculus, 2ADV C4 2012 HSC 11g

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sec^2\left(\frac{x}{2}\right) \, dx \\ &= \left[2 \tan\left(\frac{x}{2}\right)\right]_0^{\frac{\pi}{2}} \\ &= 2 \tan \frac{\pi}{4} - 2 \tan 0 \\ &= 2(1) - 0 \\ &= 2\end{aligned}$$

7. Calculus, 2ADV C4 2014 HSC 11e

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \sin\left(\frac{x}{2}\right) dx \\ &= \left[-2 \cos\left(\frac{x}{2}\right) \right]_0^{\frac{\pi}{2}} \\ &= -2 \left[\cos\left(\frac{\pi}{4}\right) - \cos 0 \right] \\ &= -2 \left[\frac{1}{\sqrt{2}} - 1 \right] \\ &= -\frac{2}{\sqrt{2}} + 2 \\ &= 2 - \sqrt{2} \end{aligned}$$

8. Calculus, 2ADV C4 2014 HSC 13a

i. $y = 3 + \sin 2x$

$$\frac{dy}{dx} = 2 \cos 2x$$

ii.
$$\int \frac{\cos 2x}{3 + \sin 2x} dx$$
$$= \frac{1}{2} \int \frac{2 \cos 2x}{3 + \sin 2x} dx$$
$$= \frac{1}{2} \ln(3 + \sin 2x) + C \quad (\text{from part (i)})$$

9. Calculus, 2ADV C4 2008 HSC 2cii

$$\begin{aligned} & \int_0^{\frac{\pi}{12}} \sec^2 3x \, dx \\ &= \left[\frac{1}{3} \tan 3x \right]_0^{\frac{\pi}{12}} \\ &= \left[\left(\frac{1}{3} \tan\left(\frac{\pi}{4}\right) \right) - \left(\frac{1}{3} \tan 0 \right) \right] \\ &= \frac{1}{3}(1) - 0 \\ &= \frac{1}{3} \end{aligned}$$

10. Calculus, 2ADV C4 2010 HSC 5b

i. Need to prove

$$\sec^2 x + \sec x \tan x = \frac{1 + \sin x}{\cos^2 x}$$

$$\begin{aligned} \text{LHS} &= \sec^2 x + \sec x \tan x \\ &= \frac{1}{\cos^2 x} + \frac{1}{\cos x} \times \frac{\sin x}{\cos x} \\ &= \frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} \\ &= \frac{1 + \sin x}{\cos^2 x} \\ &= \text{RHS} \quad \dots \text{ as required} \end{aligned}$$

ii. Need to prove

$$\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}$$

$$\text{i.e. } \frac{1 + \sin x}{\cos^2 x} = \frac{1}{1 - \sin x} \quad (\text{part (i)})$$

$$\begin{aligned} \text{LHS} &= \frac{1 + \sin x}{\cos^2 x} \\ &= \frac{1 + \sin x}{1 - \sin^2 x} \\ &= \frac{1 + \sin x}{(1 - \sin x)(1 + \sin x)} \\ &= \frac{1}{1 - \sin x} \quad \dots \text{ as required} \end{aligned}$$

$$\text{iii. } \int_0^{\frac{\pi}{4}} \frac{1}{1 - \sin x} dx$$

$$= \int_0^{\frac{\pi}{4}} (\sec^2 x + \sec x \tan x) dx$$

$$= [\tan x + \sec x]_0^{\frac{\pi}{4}}$$

$$= \left[\left(\tan\left(\frac{\pi}{4}\right) + \sec\left(\frac{\pi}{4}\right) \right) - (\tan 0 + \sec 0) \right]$$

$$\begin{aligned} &= \left[\left(1 + \frac{1}{\cos\left(\frac{\pi}{4}\right)} \right) - \left(0 + \frac{1}{\cos 0} \right) \right] \\ &= 1 + \sqrt{2} - 1 \\ &= \sqrt{2} \end{aligned}$$

♦♦ Mean mark 31%.

♦ Mean mark 37%.